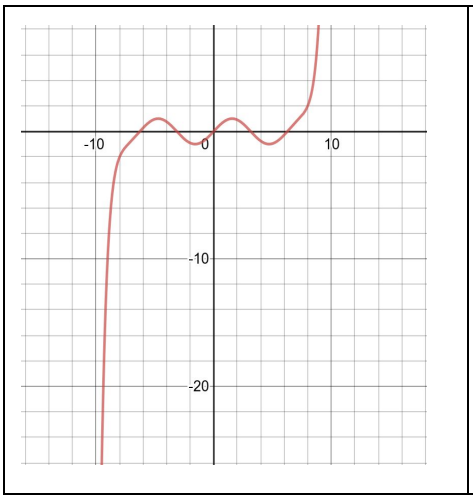
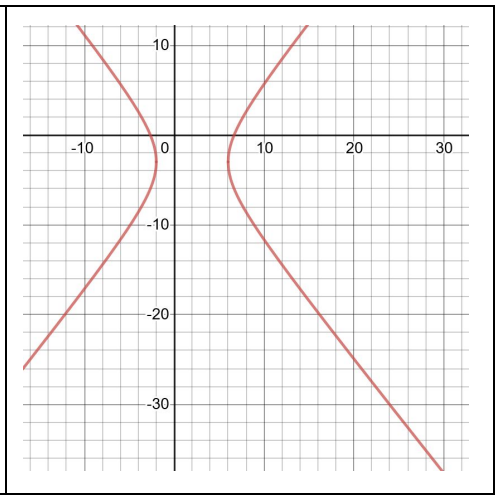
A function describes a relation between 2 sets, assigning each value in one set (called the domain; i.e. the "input" set) exactly one corresponding value in the other set (called the codomain; i.e. the "output" set). In other words, there is only one y that corresponds to one x.

To determine if a graph is a function, there is a simple method called the Vertical Line Test. You simply draw a straight line parallel to the y-axis. If there exists only 1 intersection between the line at the graph, then the graph is a function.





ABOVE:

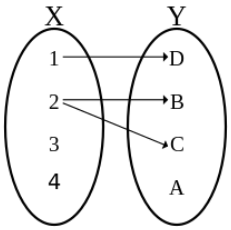
The graph on the light is a function, since for any x on the domain, you can draw a vertical line parallel to the y-axis (x = 0) and it will only intersect the graph at 1 point.

In contrast, the graph on the right is not a function, since there is at least 1 point on the domain such that if you draw a vertical line parallel to the y-axis, the line will intersect the graph at 2 distinct points. Rather, this graph is a parametric function (specifically a hyperbola), although we will not discuss that here.

On the contrary, it is possible for a function to have more than one element in its domain map to the same codomain. We'll elaborate on this further as you progress.

SIDENOTE: Codomain v. Range

The term "codomain" and "range" are not synonymous -- the codomain is often defined as the range, although some mathematicians define it as the *image*. For the sake of simplicity, we will not elaborate on this topic, as it requires some knowledge regarding sets, and will instead use the term codomain and range interchangeably.



On the diagram to the right, X represents the DOMAIN and y represents the CODOMAIN. The elements 1,2,3,4 and D,B,C,A are elements of x and y, respectively. As you can see, 1 "maps" to D, while 2 "maps" to both B and C. Since an element in the domain maps to 2 elements in the codomain, this is not a *function*, but rather a *relation*.

DOMAIN:

The domain is the set of all values that the function can take. For instance, let f(x) = x. The domain for f(x) is all real numbers ($\mathbb{R}$), since x, the input, can take any value within the set of all real numbers.

This isn't always the case, however. For example, f(x) =x cannot take any values less than 0, since on a real plane, we cannot take the square root of a negative number. Therefore, the domain of f(x) =x is [0, +∞).

The domain is typically associated with the x-axis; that is, the domain for a function is a set of values on the x-axis.

<u>RANGE:</u>
The range is the set of all outputs of the function. Let f(x) = x. The range for f(x) is all real numbers, since the output can be any number within the set of real numbers. f(x) = x^2, however, cannot output a negative number, since the square of any real number is bound to be positive. Thus, the range of f(x) = x^2 is [0, +∞). Likewise, since the square root of any number within the set [0, +∞) will be positive, the range for f(x) = $x^{½}$ is likewise [0, +∞).
The range is typically associated with the y-axis; that is, the range for a function is a set of values on the y-axis.

<u>FUNCTION TRANSFORMATION</u>

A function transformation takes a parent function *f* and transforms it in such a way through translations, reflections and dilations to create a new function.

The standard transformation form is as follows:
g(x) = ±a*f*(±b(x+h)) + k

a = vertical dilation (i.e. stretch if $|a| > 1$, compress if $|a| < 1$)

b = horizontal dilation (i.e. stretch if $|b| < 1$, compress if $|b| > 1$

h = horizontal translation (if $h > 0$, then the function is shifted *h* units to the left. If $h < 0$, then the function is shifted *h* units to the right)

k = vertical translation (if $k > 0$, then the function is shifted *k* units "up", or towards the positive y-axis. If $k < 0$, then the function is shifted *k* units "down", or towards the negative y-axis)

Additionally, if a < 0, then the function is reflected across the x-axis (y = 0). Likewise, if b < 0, then the function is reflected across the y-axis (x = 0)

f is not a variable; it simply represents the parent function *f(x).* Every transformation inside *f*(±b(x+h)) changes the x-coordinate, while
±**a***f*(±b(x+h)) + **k** (**the bolded variable**s) changes the y-coordinate.

To show transformations in action, we recommend you check out the following links below

- https://www.geogebra.org/m/DF8E4xYe

- https://www.desmos.com/calculator

END BEHAVIOR

The *end behavior* of a function describes the behavior of the graph as x approaches $\pm\infty$. In some cases, the function itself will approach $\pm\infty$, although in other cases, f(x) might approach, but never reach, a finite value (this is called an horizontal asymptote).

In Math 3, we describe end behavior as follows:
As $x \to \pm\infty$, f(x) → _____

Basically, as x approaches $\pm\infty$, f(x) will approach a certain value.

The more conventional notation (which is used in Pre-Calculus) is by writing end behavior in limit notation, or
$\lim_{x\to\pm\infty} f(x) =$ _____
Which means literally the exact same thing: As x approaches $\pm\infty$, f(x) will approach a certain value.

EVEN V. ODD FUNCTIONS
Even function **f(x) = f(-x)**
- Symmetric about the y-axis

Odd function **f(x) = f(-x)**
- Symmetric about the origin

A simple way to tell if a polynomial is an even or odd function is by looking at the exponents of the polynomial.

If all exponents are even integers (or 0, which in that case would yield a constant), then the function must be even. Likewise, if all exponents are odd integers, then the function must be odd.

Otherwise, the function is neither even nor odd.

1. *f*(x) = $2x^2 + 5$ is equivalent to $2x^2 + 5(x)^0$. Since the exponents of *f* are all even no., or else 0, *f* is an even function
2. *f*(x) = 3x^{3}+9x^{2}+4x+12. Since the exponents of *f* are not all even, or all odd, *f* is neither even nor odd.

Formally, to show that a function is even, f(x) = f(-x) for all *x* in the domain of that function.

For example:

Show that f(x) = $2x^4 + x^2$ *is even.*
We know that *f* is even if and only if *f*(x) = *f(-x).*
So if *f* is even, then $2x^4 + x^2 = 2(-x)^4 + (-x)^2$

Since a negative number raised to an even integer power will always be a positive number, we can say that
$2x^4 + x^2 = 2(-x)^4 + (-x)^2$
Therefore, f is an even function

On the other hand, to show that a function is odd, we must show that f(-x) =-f(x).

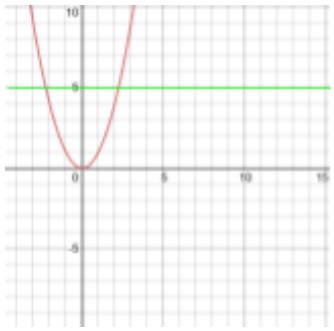
Exercise: Determine if the following functions are even, odd or neither.

1. $f(x) = x^3$
2. $f(x) = x^3 + 7$
3. $g(x) = \sin(x) * \tan(x)$

<u>1-1 FUNCTIONS (Injective):</u>

A function is 1-1 if f(a) = f(b) implies a = b. In other words, there exists only one x that corresponds to one y, or **there is exactly one input for every output**. To determine if a function is 1-1, we use the Horizontal Line Test. Draw a line parallel to the x-axis, and If for the entirety of the graph, there exists only 1 intersection between the line at the graph, then the graph is 1-1.

The inverse of a function only exists for the whole domain if that function is 1 to 1.

	This function, $f = 2^x$, is an example of a 1=1 function. We can draw a horizontal line parallel to the x-axis for the range (0, ∞) and can see that the line only intersects the function at 1 unique point.
	This parabola, although a function, is not a 1-1 function, since there are y-values that correspond to more than 1 x-value (for example, $f(-2) = f(2) = 4$). We can draw a horizontal line parallel to the x-axis, and we will see that the line will intersect the parabola at more than 1 unique point.

1-1 functions are also referred to as *injective* functions, although it is unlikely that you will need to know this term for any classes.

<u>SLOPE:</u>

The average slope given 2 points on a graph represents the average rate of change, or mathematically, the ratio between the change in y and the change in x of those two given points. Simply put, when asked to calculate the slope between 2 points *a, b*, you are asked to find the change in y over the change in x, which can be solved by the following formula:

$$m = \frac{f(b) - f(a)}{b - a}$$

Where m represents the slope, and a and b are elements in the domain (or x-values) that correspond to f(a) and f(b), respectively.

<u>LINEAR AVERAGE SLOPE:</u>

The slope of a linear function will always remain constant -- that is, the change in y over change in x for any 2 points on the graph will have a constant value. Thus, the average slope between 2 points on a linear function is the same, no matter what points you use to calculate the slope.

For example, let's look at the function $f(x) = 2x + 3$

Let's say we wanted to calculate the slope from x = -3 to x = 3.

Using the formula for slope, we see that

$$m = \frac{f(3) - f(-3)}{3 - (-3)} = \frac{9 - (-3)}{3 - (-3)} = \frac{12}{6} = 2$$

So the average slope of *f* from x = -3 to x = 3 is 2

Let's say we try this for - say, x = 2 to x = 7

$$m = \frac{f(7) - f(2)}{7 - 2}$$

$$m = \frac{17 - 7}{7 - 2} = \frac{10}{5} = 2$$

Notice how this is the same slope we calculated using the other 2 points.

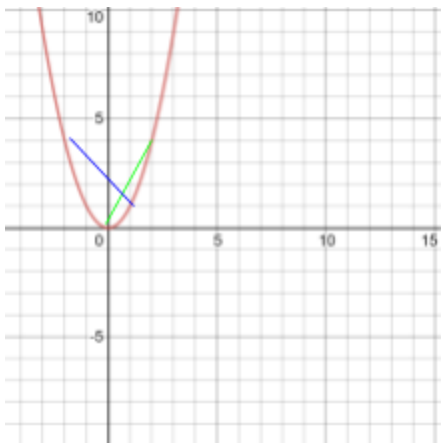
To show this is true for all $f(x)$, we can calculate the slope using variables *a,b* as placeholders

$$m = \frac{(2b + 3) - (2a + 3)}{b - a} = \frac{2b + 3 - 2a - 3}{b - a} = \frac{2(b - a)}{(b - a)} = 2$$

For any $a, b \in \mathrm{Dom}(f)$, the slope between $(a, f(a))$ *and* $(b, f(b))$ is 2

NONLINEAR AVERAGE SLOPE:

The average slope of a nonlinear function however, may vary depending on which points you use to calculate the slope



Above:

The green line represents the average slope from (0, $f(0)$) to (2, $f(2)$).

The blue line represents the average slope from (-2, $f(-2)$) to (1, $f(1)$).

For example, let's take a look at the parent quadratic function , $f(x) = x^2$

The average slope from (0, $f(0)$) to - say, (2, $f(2)$) is calculated as follows:

$m = \frac{4-0}{2} = 2$

However, say we choose to calculate the average slope from (0, $f(0)$) to (3, $f(3)$).

$m = \frac{9-0}{3} = 3$

Or from (-2, $f(-2)$) to (1, $f(1)$).

$m = \frac{1-4}{1-(-2)} = -1$

We can see that on a curve, the average slope between 2 points is subject to change, depending on which points you use and what average slope you're trying to find.

The average slope is what some people call a *secant line*: in short, it is a straight line that intersects 2 points on a graph.

Basically, average slope is a straight line connecting 2 points, and represents the average rate of change (i.e. slope) between those said 2 points.

Now there is a way to calculate the rate of change at 1 unique point on a graph, or the rate of change of a function at a particular *instant* - the *instantaneous slope,* for lack of a better term. But this requires calculus, and is actually one of the primary reasons why single-variable calculus was invented in the first place..

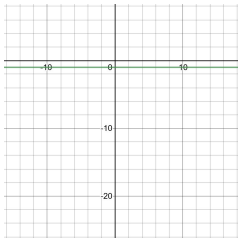
TYPES OF FUNCTIONS:

We will be briefly reviewing the **continuous** functions that you may come across in Math 1 to Pre-Calculus (AP Calculus students should be familiar with all of the functions below; if not, we advise you to review them).

NOTE:

- The domain, range and end behavior of the functions below refer to the parent function. Of course, transforming the function would change the function behavior.

- If the function doesn't have a clear "parent function", we gave an example of such a function and stated the domain, range and end behavior accordingly.

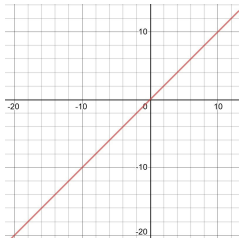


Constant

f(x) = c;
where "c" is a constant.

Dom(f) = $\mathbb{R}$
Rng(f) = c

End Behavior
$\lim_{x\to+\infty} f(x) = c$
$\lim_{x\to-\infty} f(x) = c$



Point-Slope Form:
$(Y_2 - y_1) = m\,(x_2 - x_1)$
where m is the slope and y_1 and x_1 represent a known coordinate (x_1, y_1) on the function.

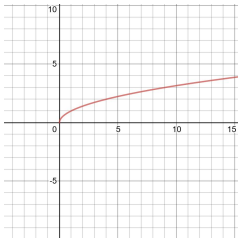
Slope-Intercept Form:
$y = mx + b$
where m is the slope and b is the y intercept (the point where the function crosses the y-axis)

Linear

Parent Function
f(x) = x

Dom(f) = $\mathbb{R}$
Rng(f) = $\mathbb{R}$

End Behavior
$\lim_{x\to+\infty} f(x) = \pm\infty$
$\lim_{x\to-\infty} f(x) = \mp\infty$



Standard Form:
f(x) = $\pm a\sqrt{\pm b(x+h)}$ +k

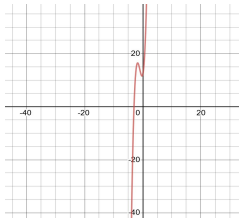
Square Root

Parent Function
f(x) = $\sqrt{x}$

Dom (f) = $[0, +\infty)$

Rng (f) = $[0, +\infty)$

End Behavior
$\lim_{x\to+\infty} f(x) = \infty$
$\lim_{x\to0+} f(x) = 0$



f(x) = 3x^{3}+9x^{2}+4x+12

Standard Form:
f(x) = $\pm a\,(\pm b(x+h))^n$ +k

n - highest degree

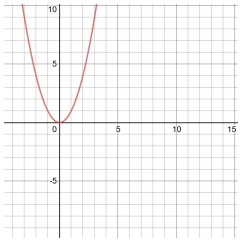
Polynomial

If *f(x)* is an even function; that is, f(x) = f(-x),
Dom(f) = $\mathbb{R}$
Rng (f) =
$[c, +\infty)$ if f(x) = c is the absolute min. value of the function
$(-\infty, c]$ if f(x) = c is the absolute max. value of the function

End Behavior
$\lim_{x\to+\infty} f(x) = \pm\infty$
$\lim_{x\to-\infty} f(x) = \pm\infty$

If *f(x)* is an odd function; that is,
f(-x) = -f(x),
Dom(f) = $\mathbb{R}$
Rng (f) = $\mathbb{R}$

$\lim_{x\to+\infty} f(x) = \pm\infty$
$\lim_{x\to-\infty} f(x) = \mp\infty$



Standard Form:
$f(x) = ax^2 + bx + c$

OR:

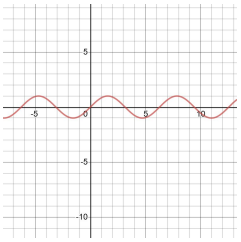
$f(x) = \pm a\,(\pm b(x+h))^2 + k$
(i.e. vertex form)

Quadratic

<u>Parent Function</u>
$f(x) = x^2$

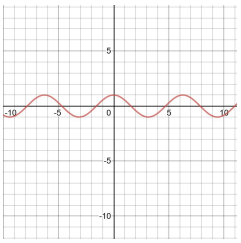
Dom (f) = $\mathbb{R}$
Rng (f) = $[0, +\infty)$

<u>End Behavior</u>
$\lim_{x\to+\infty} f(x) = \infty$
$\lim_{x\to-\infty} f(x) = \infty$



Standard Form:
$f(x) = \pm a \sin(\pm b(x+h) + k$
where
a = amplitude
b = frequency
h = phase shift
k = vertical shift

The period of a sinusoidal function is equal to $\frac{2\pi}{b}$, where b is the frequency.



Standard Form:
$\pm a \cos(\pm b(x+h) + k$

Sinusoidal (sin(x); cos(x))

<u>Parent Function</u>
$f(x) = \sin(x)$

Dom(f) = $\mathbb{R}$
Rng (f) = $[-1, 1]$

<u>End Behavior</u>
$\lim_{x\to+\infty} f(x)$ = diverges by oscillation
$\lim_{x\to-\infty} f(x)$ = diverges by oscillation

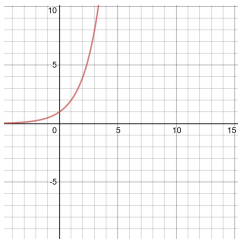
<u>Parent Function</u>
$f(x) = \cos(x)$

Dom(f) = $\mathbb{R}$
Rng (f) = $[-1, 1]$

<u>End Behavior</u>
$\lim_{x\to+\infty} f(x)$ = diverges by oscillation
$\lim_{x\to-\infty} f(x)$ = diverges by oscillation

Observe that:
$\sin(x + \frac{\pi}{2}) = \cos(x)$
$\cos(x + \frac{\pi}{2}) = -\sin(x)$

We will elaborate upon sinusoidal functions in our trigonometry section.



$f(x) = 2^x$

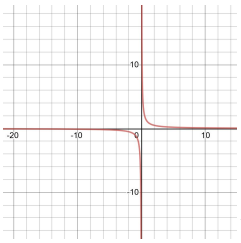
Standard Form:
$f(x) = \pm a\,(B)^{\pm(b(x-h))} + k$
B - base

Exponential

<u>Parent Function</u>
$f(x) = B^x$

Dom (f) = $\mathbb{R}$
Rng (f) = $(0, +\infty)$

<u>End Behavior</u>
$\lim_{x\to+\infty} f(x) = \infty$
$\lim_{x\to-\infty} f(x) = 0$



When asked to graph rational functions, here are some things you should look for:

- Points of Discontinuity (i.e. "Holes")

Rational

<u>Parent Function</u>
$f(x) = \frac{1}{x}$
Dom (f) = $(-\infty, 0) \cup (0, +\infty)$
Rng (f) = $(-\infty, 0) \cup (0, +\infty)$

<u>End Behavior</u>
$\lim_{x\to+\infty} f(x) = 0$
$\lim_{x\to-\infty} f(x) = 0$

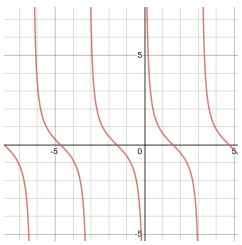
- Vertical Asymptote
- Horizontal Asymptote
- x - intercept
- y - intercept
- End Behavior
- Vertical Asymptotic Behavior:

We will explain how to find these values in a later section.

Vertical Asymptotic Behavior:

$\lim_{x \to 0+} f(x) = +\infty$

$\lim_{x \to -0-} f(x) = -\infty$



Standard Form:

$f(x) = \pm a \cot(\pm b(x + h) + k$

The period of a cotangent function is equal to $\frac{\pi}{b}$, where b is the frequency.

Cotangent

$\cot(x) = \frac{1}{\tan(x)}$

<u>Parent Function</u>

$f(x) = \cot(x)$

Dom (f) = $\mathbb{R}$, $x \neq \pi k$,

Where k is an integer

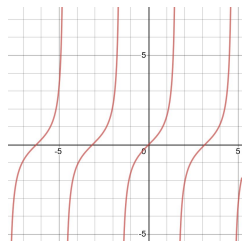
Rng (f) = $\mathbb{R}$

<u>End Behavior</u>

for $0 < x < \pi$,

$\lim_{x \to \pi-} f(x) = -\infty$

$\lim_{x \to -0+} f(x) = +\infty$



Standard Form:

$f(x) = \pm a \tan(\pm b(x + h) + k$

The period of a tangent function is equal to $\frac{\pi}{b}$, where b is the frequency.

Tangent

<u>Parent Function</u>

$f(x) = \tan(x)$

Dom (f) = $\mathbb{R}$, $x \neq \frac{\pi}{2}k$, where k is an integer

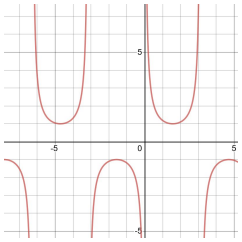
Rng (f) = $\mathbb{R}$

<u>End Behavior</u>

for $-\frac{\pi}{2} < x < \frac{\pi}{2}$,

$\lim_{x \to \frac{\pi}{2}-} f(x) = +\infty$

$\lim_{x \to \frac{-\pi}{2}+} f(x) = -\infty$



Standard Form:
f(x) = $\pm a\ csc\ (\pm b(x+h)$ +k

The period of a cosecant function is equal to $\frac{2\pi}{b}$, where b is the frequency.

Cosecant
csc(x) = $\frac{1}{sin(x)}$

<u>Parent Function</u>
f(x) = csc(x)

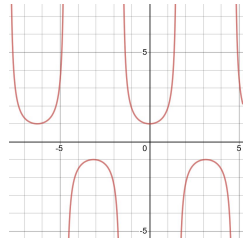
Dom (f) = ℝ, x ≠ πk
,
where k is an integer

Rng (f) = (-∞, -1] ∪ [1, +∞)

<u>End Behavior</u>
for 0 < x < π,

$\lim_{x\to\pi-}$ f(x) = +∞
$\lim_{x\to-0+}$ f(x) = +∞

for 0 < x < π,
$\lim_{x\to2\pi-}$ f(x) = -∞
$\lim_{x\to-0+}$ f(x) = -∞



Standard Form:
f(x) = $\pm a\ sec(\pm b(x+h)$ +k

The period of a secant function is equal to $\frac{2\pi}{b}$, where b is the frequency.

Secant
sec(x) = $\frac{1}{cos(x)}$

<u>Parent Function</u>
f(x) = sec(x)

Dom (f) = ℝ, x ≠ $\frac{\pi}{2}k$, where k is an integer

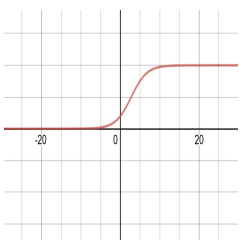
Rng (f) = (-∞, -1] ∪ [1, +∞)

<u>End Behavior</u>
for - $\frac{\pi}{2}$ < x < $\frac{\pi}{2}$,

$\lim_{x\to\pi-}$ f(x) = +∞
$\lim_{x\to-0+}$ f(x) = +∞

for $\frac{\pi}{2}$ < x < $\frac{3\pi}{2}$,

$\lim_{x\to2\pi-}$ f(x) = -∞
$\lim_{x\to-0+}$ f(x) = -∞



$f(x) = \frac{10}{1+4e^{-.5x}}$

Dom (f) = ℝ
Rng (f) = (0, L)

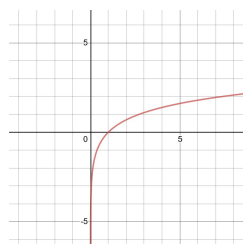
<u>End Behavior</u>
$\lim_{x\to+\infty}$ f(x) = C
$\lim_{x\to-\infty}$ f(x) = 0

Logistic:

<u>Pre-Calculus</u>
f(x) = $\frac{L}{1+ae^{-kx}}$
where
L = Limit to growth (horizontal asymptote)

a, k = constants
(0, $\frac{L}{1+a}$) = y-intercept

<u>Calculus</u>
The rate of change of a logistic function is a function of f(x) rather than x. You can think of this as the rate of change being dependent on



$f(x) = \ln(x)$

Dom (f) = [0,+∞)
Rng (f) = ℝ

<u>End Behavior</u>
$\lim_{x\to+\infty}$ f(x) = +∞
$\lim_{x\to-0+}$ f(x) = -∞

Logarithmic

$f(x) = \log_b(x)$
where
b = Base of logarithm

NOTE:
$\log_e(x)$, where e is Euler's constant, is often abbreviated as *ln(x)*

ln(x) is called the *natural logarithm*

$\log_{10}(x)$, is often referred to as the *common logarithm.*

Again, we will discuss this further in a later section

the population, as that is what logistic models are usually applicable for.

$$\frac{dy}{dx} = ky\,(L - y)$$

Solving this differential equation yields:

$$y = \frac{L}{1+ae^{-kLx}}$$

where

L = Limit to growth (horizontal asymptote)

$a \in \mathbb{R}$

K = rate of growth

e = Euler's constant (≈2.7128)